

CS 370 – Assignment 2 Report

Files Modified

The following files were modified: - `kernel/proc.h` - `kernel/proc.c` - `kernel/sysproc.c` - `kernel/syscall.c` - `kernel/syscall.h` - `kernel/defs.h` - `user/user.h` - `user/usys.pl` - `user/evaluate.c` - `Makefile` - `kernel/trap.c`

Hypothesis

I hypothesize that our Fairly Fair Scheduler algorithm will improve the turnaround time of higher-rudeness processes by allocating them a larger share of CPU time. Processes with higher rudeness values get proportionally larger time slices, so they should finish CPU-intensive work faster than those with lower rudeness values. The evaluation metric is turnaround time, measured in scheduler ticks, which reflects the scheduler's impact on process completion time.

Measured Data

Run	Rudeness 5 (Sum)	Rudeness 15 (Normalize)	Rudeness 30 (Bubble Sort)	Total Time
1	0	0	220	223
2	0	0	219	221
3	0	0	222	225
4	0	1	218	221
5	0	0	218	221
6	0	0	219	222

All values are measured in scheduler ticks.

Results

The results partially support the hypothesis. While the bubble sort process required approximately 218–222 ticks to complete, the sum and normalization tasks completed in 0–1 ticks since these were much smaller tasks. In conclusion, the longer-running bubble sort task best demonstrates the scheduler's behavior, while the lighter tasks do not run long enough to clearly show the impact of the different rudeness values tested.